

■ 3.7.9 Advanced Topic: Generalized Linear Algebra

The matrix operations discussed in the previous section work by trying to find combinations of matrix elements which give zero. By default, a simple test of the form `e==0` is effectively applied to each combination that is generated to see if it is in fact zero.

Sometimes you may want to apply more complicated tests to determine whether combinations of matrix elements should be considered to be zero. The option **ZeroTest** allows you to specify a function which is applied to each combination that is generated to determine whether the combination should be considered to be zero.

```
LinearSolve[m, b, ZeroTest -> f], etc.
```

apply *f* to determine whether combinations of matrix elements are zero

Generalized linear algebra.

Here is a 2×2 matrix.

```
In[1]:= m = {{1, 1, 0}, {0, 1, 1}, {1, 0, 1}}
```

```
Out[1]= {{1, 1, 0}, {0, 1, 1}, {1, 0, 1}}
```

In the usual case, `m` has no null space.

```
In[2]:= NullSpace[m]
```

```
Out[2]= {}
```

If, however, combinations of matrix elements need only be zero modulo two, `m` does have a non-trivial null space.

```
In[3]:= NullSpace[m, ZeroTest -> (Mod[#, 2]==0 &)]
```

```
Out[3]= {{1, -1, 1}}
```

You can use **ZeroTest** to do linear algebra with matrices whose elements are not necessarily ordinary real numbers. One important application is in working with matrices whose elements are symbolic expressions. When complicated symbolic expressions are involved, it may be necessary to apply various transformations in order to find out whether particular combinations of expressions are in fact zero. You can use **ZeroTest** to specify such transformations.